

- Module Objectives
- Rail Fence Cipher
- Permutation Cipher
- Product Cipher
- Video: Product Cipher
- 1 min
- Reading: Lecture Slides for Classical Cipher - Transposition
- 13 min
- Quiz: Product Cipher
- Submitted

● Congratulations! You passed!

Grade received 100%

Latest Submission Grade 100%

Product Cipher

Go to next item

Review Learning Objectives

1. The plaintext MEETMELATER gets processed by a product cipher, comprised of Caesar Cipher (with a key of 23) and Permutation Cipher (with a key of [5 3 1 4 2] and no padding with extra letters). What is the corresponding ciphertext?
- 4 / 4 points

Submit your assignment

Due Apr 14, 11:59 PM IST Attempts 3 every 8 hours

Try again

EXBBBIOQJBO

Receive grade

Your grade

100%

View Feedback

We keep your highest score

2. The same product cipher in the previous question comprised of Caesar Cipher and Permutation Cipher (with a key of length 5), is again used but with different keys. Given a plaintext, how many distinct ciphertexts can there be?
- 4 / 4 points

3120

Correct

This is equal to the product between the key size of the Caesar Cipher and the key size of Permutation Cipher.

3. The same product cipher in the previous question, comprised of Caesar Cipher and Permutation Cipher (with a key of length 5), is used to encrypt a long English book (ignoring capitalization and other symbols beyond English letters). Your friend makes the following claims. Select all those that are correct.
- 6 / 6 points

☐ "If we another Caesar Cipher encryption in the product cipher (so that there are two Caesar Cipher operations and a Permutation Cipher operation), then it becomes harder to crack by an attacker (who does not know the key)."

☒ "Letter Z appeared the most in the ciphertext, so it must be mapped from the latter E."

Correct

☒ "If we replace the Caesar Cipher with a Monoalphabetic Cipher, then it becomes harder to crack by an attacker (who does not know the key)."

Correct

☐ "The letter-pair/digram AX appeared the most in the ciphertext, so they must be TH. In other words, A is mapped from the letter T, and X is mapped from H."

☒ "The ordering between the Caesar and the Permutation does not matter within the product cipher, so we can decrypt it in any order."

Correct